

PIYUSH NAIK

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PROFESSIONAL SUMMARY

Final-year B.Tech (Artificial Intelligence & Machine Learning) student with hands-on experience building deep learning applications, including deepfake detection systems using EfficientNet, PyTorch, and Grad-CAM explainability, along with data analytics pipelines in Python and SQL. Seeking an entry-level Machine Learning Engineer / Software Engineer role to apply strong analytical and applied AI skills to real-world problems.

EDUCATION

Yeshwantrao Chavan College of Engineering (YCCE), Nagpur
B.Tech in Artificial Intelligence & Machine Learning

Expected Graduation: 2027

TECHNICAL SKILLS

Programming Languages: Python, SQL

Machine Learning & Deep Learning: PyTorch, scikit-learn, Convolutional Neural Networks (CNN), EfficientNet

Computer Vision & Explainability: MTCNN Face Detection, Grad-CAM

Data Analysis & Visualization: NumPy, Pandas, Matplotlib

Databases: MySQL

Tools & Platforms: Git, GitHub, VS Code, Streamlit, MySQL Workbench

Currently Learning: React.js

PROJECTS

AI Video Detector (Deepfake Detection) | Python, PyTorch, EfficientNet-B0, MTCNN, Grad-CAM, Streamlit

- Developed a deep learning system that analyzes uploaded videos and classifies them as AI-generated (deepfake) or real, using EfficientNet-B0 trained on a Kaggle dataset.
- Built a per-frame pipeline with MTCNN face detection (center-crop fallback), 224x224 resizing, ImageNet normalization, and softmax probabilities for Real vs AI.
- Designed confidence-weighted aggregation of frame scores into a video-level prediction so uncertain frames do not dilute strong evidence.
- Added Grad-CAM explainability and timestamp evidence showing the frames with the strongest AI and real predictions.
- Built an interactive Streamlit web interface for uploading videos and viewing results.

AI Image Detector (Deepfake Detection) | Python, PyTorch, Deep Learning

- Built a deep learning image classifier that distinguishes AI-generated (deepfake) images from real ones, trained on a Kaggle dataset.

eSports Tactical Match & Telemetry Analyzer | MySQL, Python, Pandas, NumPy, Matplotlib

- Designed a normalized MySQL schema with 4 relational tables linked via foreign keys to model match, player, and round-level data.
- Implemented SQL window functions (LAG, LEAD, DENSE_RANK) to compute round-over-round momentum shifts and player efficiency rankings.
- Built a Python data pipeline with Pandas and NumPy to clean data, handle NULL values, and compute custom metrics (K/D ratio, efficiency rating).
- Developed a multi-panel Matplotlib dashboard visualizing win rates, economy impact, and player performance trends across matches.

Netflix Content Trends Analysis | Python, Pandas, Matplotlib

- Analyzed 8,800+ Netflix titles and built Matplotlib visualizations of content type, ratings, regional production, and movie duration trends.
- Identified a sharp rise in title additions after 2015 and a shift toward TV Show production for viewer retention.
- Ranked top content-producing countries and presented findings as a data storytelling narrative for non-technical audiences.

CERTIFICATIONS

- Machine Learning Specialization, Coursera
- Software Engineering Job Simulation, JPMorgan Chase & Co. (Forage)
- Data Analyst Job Simulation, Deloitte (Forage)